SAHIL BARAK

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PORTFOLIO | **GITHUB** | **LEETCODE**

SUMMARY

Full-stack developer with B.Tech and M.Tech in Computer Science and Engineering. Builds production-grade products with Next.js, TypeScript, Node.js, and PostgreSQL, and has experience with microservices, real-time systems, and cloud deployment. Seeking a full-stack developer role where I can build scalable web and mobile products.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, HTML5, CSS3, SQL
Frontend: React, Next.js, React Native, Tailwind CSS, Framer Motion, XState
Backend: Node.js, Express, REST APIs, WebSockets, JWT Authentication, Hocuspocus
Databases and Real-Time: PostgreSQL, Redis, MongoDB, Prisma ORM, Drizzle ORM, Yjs (CRDTs)
DevOps: Docker, Git, GitHub Actions (CI/CD), Vercel, Cloudflare, AWS

PROJECTS

CANVASFLOW – REAL-TIME COLLABORATIVE WHITEBOARD

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TECHNOLOGIES **-** *TypeScript, Next.js, NestJS, React, Yjs, WebSockets, PostgreSQL, Redis, Docker, Cloudflare R2, Fly.io*

- Built a full-stack collaborative whiteboard in a TypeScript monorepo with Next.js, React, NestJS, WebSockets sync, claude API diagram and polar billing. Designed image storage with cloudflare R2 with presigned direct uploads.
- Engineered a custom canvas engine with R-tree indexing, connected/elbow arrows and auto-layout mind maps, achieving 59 FPS with 2,000 shapes.
- Built live collaboration over WebSockets with Yjs CRDTs: co-editing, presence, offline editing and canvas comments with @mentions. Load-tested 200 users per board with zero lost updates, reduced crash recovery from 23.9 s to 4.1 s.
- Secured the platform with JWT/OAuth, role-based access, rate limits on 89 of 94 routes, CSP/HSTS and CodeQL, bringing critical vulnerabilities to 0. Raised test coverage to 88% on the sync server (from 0%) and 85% on auth (from 11%), with 89/89 end-to-end tests passing.

MOX UI – COMPONENT LIBRARY WEBSITE

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TECHNOLOGIES **-** *TypeScript, Next.js, Tailwind CSS, Framer Motion, Shadcn*

- Built a UI component library website inspired by 21st.dev and Aceternity UI, with a browsable collection of reusable components.
- Designed the site around a component registry pattern, so each component can be added and maintained independently.
- Added smooth UI animations with Framer Motion and syntax-highlighted code previews with Shiki.

EDUCATION

M.Tech, Computer Science and Engineering [CGPA – 8.3] **2024 - 2026**
Deenbandhu Chhotu Ram University of Science and Technology (DCRUST) Murthal, Haryana

B.Tech, Computer Science and Engineering [CGPA – 7.31] **2019 - 2023**
PDM University Bahadurgarh, Haryana

CERTIFICATIONS & EXTRACURRICULAR ACTIVITIES

- **Google** UI/UX Design Professional Certificate
- **GATE 2024:** All India Rank 1291 (Computer Science and Information Technology).
- Led the college cricket team at the Inter-university tournament, 2022.

HOBBIES

- Playing weekly corporate cricket
- Gym training and fitness
- Traveling and exploring new places